VT-447V(I)

Laminate/Prepreg

UL Approval: E214381 Version: 29/10/2025 /127 /128 /130 /154

General Information

- Phenolic Cured System
- Halogen free
- High Td & High Tg (190°C)
- FR15.1 & MOT 150°C
- CAF Resistance
- Low Z-CTE
- Improved Mechanical Process-ability & Excellent Thermal Reliability

Application

Mobile Phone, Smart Phone, Automotive, High Power Application, Server, Computer, Communication Equipment, Instrumentation, Electronic Game Machine, VCR, etc.

Availability

- Core Thickness: 0.002" (0.05mm) to 0.200" (5mm), available in sheet or panel form
- Copper Foil: 1/4oz to 12oz
- Prepregs are available in roll or panel form
- E-Glass styles: 7628, 1506, 1500, 2113, 2313, 3313, 2116, 1080, 1086, 1078, 106 &1067, etc

Note: For cores ≤ 0.005", it is recommended to use the reverse treated copper due to the low profile. The peel strength for RT foil is ≈1-2lbs/in (0.35Kg/m) less than standard foil.

Storage Condition

Properties		Prepreg		Laminate
Storage Condition	Temperature	Below 23°C (73°F)	Below 5°C (41°F)	Room
	Relative humidity	Below 55% RH	/	/
Shelf Life		3 months	6 months	24 months (airproof)

Note: The prepreg exceeding shelf life should be retested.

VT-447V(I)

Laminate/Prepreg

UL Approval: E214381 Version: 29/10/2025 /127 /128 /130 /154

IPC-4101E WAM1 /127 /128 /130 /154

Properties		Test Method	Units	Specification	Typical Value
Thermal Properties					
Tg	DSC	IPC-TM-650 2.4.25	°C	–	190
	TMA	IPC-TM-650 2.4.24	°C	170 minimum	180
Td		ASTM D3850	°C	340 minimum	390
T260		IPC-TM-650 2.4.24.1	Minute	30 minimum	>60
T288		IPC-TM-650 2.4.24.1	Minute	15 minimum	>60
Thermal Stress @ 288°C		IPC-TM-650 2.4.13.1	Second	Pass 10s	>600
Z-axis CTE	Before Tg	IPC-TM-650 2.4.24	ppm/°C	60 maximum	30~40
	After Tg	IPC-TM-650 2.4.24	ppm/°C	300 maximum	140~160
	Total Expansion (50~260°C)	IPC-TM-650 2.4.24	%	3.0 maximum	1.8
X-Y CTE		IPC-TM-650 2.4.24	ppm/°C	–	11~13
MOT		UL 746E	°C	–	150
Electrical Properties					
DK @ 1GHz RC 50%		IPC-TM-650 2.5.5.9	–	5.4 maximum	4.40
DF @ 1GHz RC 50%		IPC-TM-650 2.5.5.9	–	0.035 maximum	0.012
Volume Resistivity	After Moisture Resistance	IPC-TM-650 2.5.17.1	MΩ-cm	1.0E+4 minimum	5.0E+8
	E-24/125	IPC-TM-650 2.5.17.1	MΩ-cm	1.0E+3 minimum	5.0E+6
Surface Resistivity	After Moisture Resistance	IPC-TM-650 2.5.17.1	MΩ	1.0E+4 minimum	5.0E+7
	E-24/125	IPC-TM-650 2.5.17.1	MΩ	1.0E+3 minimum	5.0E+6
Electrical Strength		IPC-TM-650 2.5.6.2	Volt/mil (KV/mm)	762 (30) minimum	1200~1400 (54)
Dielectric Breakdown		IPC-TM-650 2.5.6	KV	40 minimum	60
Comparative Tracking Index (CTI)		ASTM D3638	Rating (Volt)	–	Grade 3 (175-250)
Arc Resistance		ASTM D495	Second	–	195
Mechanical Properties					
Peel Strength (1oz)	As received	IPC-TM-650 2.4.8	lb/in (N/mm)	–	6~8 (1.05~1.4)
	After thermal stress	IPC-TM-650 2.4.8	lb/in (N/mm)	6 (1.05) minimum	6~8 (1.05~1.4)
Flexural Strength	Warp	IPC-TM-650 2.4.4	KPsi (MPa)	60 (415) minimum	87 (600)
	Fill	IPC-TM-650 2.4.4	KPsi (MPa)	50 (345) minimum	77 (530)
Physical Properties					
Moisture Absorption		IPC-TM-650 2.6.2.1	%	0.80 maximum	0.12
Flammability		UL-94	Rating	V-0 minimum	V-0

Note: All test data provided are typical values and are not intended to be specification values.